

1 Requirement identification

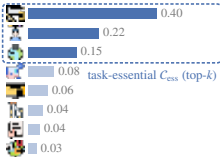
Task card D_τ^{spec}

- task description
- input/output format
- evaluation target
- a few exemplars

Identifier g_ϕ

Transformer encoder
two-layer MLP
softmax

requirement $\mathbf{r}_\tau \in \Delta^{\text{ICI}}$



Offline pretraining of g_ϕ (once, shared by all tasks)

$\mathbf{w}^{(m)} \sim \text{Dir} \rightarrow \text{probe distill} \rightarrow (\Delta \mathbf{s}^{(m)}, \Delta U^{(m)})$

$$\min_{\phi} \sum_{\tau, m} (\Delta U_{\tau}^{(m)} - g_{\phi}(D_{\tau}^{\text{spec}})^{\top} \Delta \mathbf{s}_{\tau}^{(m)})^2 + \lambda_{\text{ent}} \mathcal{H}(g_{\phi})$$

2 Targeted generation and distillation

next step: $\mathbf{w}_{t+1}, b_{t+1}$

allocation $\mathbf{w}_t$



Template library $\mathcal{P}_c$

"[N] pens cost [P]; cost of [M]?"
typed slots · fixed seeds · format rules
difficulty $d_c(x)$
easy medium hard
curriculum over t

prompt x

"3 pens cost \$4.50;
how much are 7?"



Teacher T
70B, frozen

answer y

"\$1.50 each,
so \$10.50"

$$\mathcal{L}_{\text{distill}} = -\sum_j \log p_{\theta_i}(y_j | x, y_{<j})$$

token-level distillation on (x, y)

token budget B



Student S_t
8B, updated
 $\rightarrow S_{t+1}$

S_{t+1}

3 Uncertainty-aware allocation

contextual bandit

context $\mathbf{x}_t = [\mathbf{r}_\tau; \mathbf{s}^{\text{probe}}(S_t); b_t; \rho_t]$

$h_{\eta_1}, \dots, h_{\eta_J} \rightarrow \mu(\mathbf{x}_t, \mathbf{w})$
 $\sigma(\mathbf{x}_t, \mathbf{w})$

candidates $\mathcal{A}(\tau)$

bootstrap MLP ensemble



$$\mathbf{w}_{t+1} = \arg\max_{\mathbf{w} \in \mathcal{A}(\tau)} \mu(\mathbf{x}_{t+1}, \mathbf{w}) + \kappa \sigma(\mathbf{x}_{t+1}, \mathbf{w})$$

append $(\mathbf{x}_t, \mathbf{w}_t, \hat{R}_t)$

proxy reward

$$\hat{R}_t = \mathbf{r}_\tau^{\top} \Delta \mathbf{s}_t - \beta \text{Spill}_t - \lambda \text{cost}_t$$

$$\text{Spill}_t = \sum_{c \in \mathcal{C}_{\text{ess}}} r_{\tau, c} [-\Delta s_{t, c}]_+$$

probe suite

☒ ☒ ☒ ☐
small, fixed

$$\Delta \mathbf{s}_t = \mathbf{s}^{\text{probe}}(S_{t+1}) - \mathbf{s}^{\text{probe}}(S_t)$$

